

Accelerating Quality Assurance through PACE Automation Framework for a Global Asset Management Firm

Introduction

A leading U.S.-based asset management firm, operating across 45 cities in 22 countries, serves institutional, private, and retail clients with a strong focus on research-driven investment strategies. Their technology ecosystem powers complex portfolio management, quantitative analytics, and equity trading operations; where speed, precision, and reliability are paramount.

To ensure stability across mission-critical trading and portfolio applications, the firm partnered with NuWare to reimagine their testing and quality assurance (QA) approach. The objective: to transition from lengthy manual testing cycles to a high-performance, automated regression framework capable of keeping pace with frequent releases and cross-application dependencies

Leveraging NuWare's proprietary PACE (Platform for Automation Control and Execution) framework, the client achieved enterprise-grade automation efficiency, delivering faster releases, stronger quality assurance, and measurable cost savings.

The Challenge

The firm's equity portfolio management and trading suite consisted of multiple integrated applications, each developed using different technologies and connected to several accounting systems. While comprehensive, this ecosystem created significant QA challenges:

1. **High Regression Volume**

Over **3,500 test cases** spanned across applications, making end-to-end testing highly time-intensive.

2. **Limited Testing Bandwidth**

A single manual regression cycle took **44 business days**, making it nearly impossible to sustain the client's **bi-monthly release cadence**.

3. **Defects Leaking into Production**

Insufficient QA turnaround led to defect leakage, increasing production risk and post-release support costs.

4. **Fragmented Technology Landscape**

Each application was built on different technologies, making single-tool automation infeasible.

5. **Previous Automation Failures**

Earlier automation attempts had failed due to siloed frameworks, lack of integration between tools, and limited scalability.

6. **Data Sensitivity and Time Dependency**

Test data was dynamic and interdependent; with overnight data refreshes requiring multi-day test continuity and synchronization.

The client needed a unified automation solution capable of handling complex, multi-system dependencies, dynamic data conditions, and integration with existing test management tools, without disrupting ongoing release schedules

NuWare's Approach

After a detailed feasibility study, NuWare recommended its proprietary PACE Automation Framework, an enterprise-ready, tool-agnostic platform designed to orchestrate automation across multiple technologies, environments, and timelines.

PACE offered the flexibility, intelligence, and scalability required for this client's unique testing landscape.

Key Capabilities Leveraged

- **Concurrent Execution:** Ability to run tests in parallel across multiple desktops, reducing execution time dramatically.
- **Modular Keyword-Driven Design:** Broke large test cases into **reusable “business components” (keywords)** that could be executed independently or in combination across applications.
- **Multi-Tool Compatibility:** Supported integration of different automation tools within a single test flow, ensuring complete end-to-end coverage.
- **Suspended and Resumable Execution:** Allowed long-running tests to pause and resume based on **specific events or time triggers**, critical for overnight data refresh cycles.
- **Centralized Reporting:** Real-time, unified reporting provided visibility into execution progress and outcomes across distributed test environments.
- **Dynamic Test Data Management:**
 - Integrated both **static and real-time datasets** directly from application databases.

- Enabled parameterization at multiple levels; test case, keyword, and test set, ensuring flexibility and accuracy.
- **Test Management Integration:** PACE integrated seamlessly with existing **ALM tools** for traceability, scheduling, and defect tracking.

Together, these capabilities allowed the client to run automated tests across heterogeneous environments and applications as part of a single, orchestrated regression suite.

Outcomes

The PACE-driven automation solution delivered measurable performance, cost, and quality benefits within the first year of deployment:

1. Drastic Reduction in Test Cycle Time

Regression execution time was reduced from **44 business days to under 6 days**, enabling the client to support two releases per month effortlessly.

2. High ROI and Cost Savings

- Manual regression cost: **\$8,800 per cycle**
- Savings per automated cycle: **\$8,000**
- Automation investment: **\$100,000**
- **ROI achieved within 12 regression cycles**, or less than one year

3. Enhanced Test Coverage and Reliability

- Achieved **comprehensive test coverage** across 3,500+ cases, including multi-system workflows.
- Improved consistency and defect detection before production deployment.

4. Increased Efficiency and Stability

- Unattended, concurrent executions reduced resource dependency.
- Intelligent error handling ensured smooth, continuous execution.

5. Seamless Integration Across Technologies

- Unified multiple testing tools and systems without redevelopment.
- Improved visibility through centralized dashboards and real-time analytics.

Future Outlook

Following the success of PACE's implementation, NuWare is partnering with the client to advance toward **next-generation intelligent automation**, focusing on:

- **AI-Augmented Test Case Selection:** Predictive analytics to prioritize regression tests based on risk and code impact.
- **Continuous Testing in CI/CD Pipelines:** Integrating PACE with Jenkins and Azure DevOps for real-time feedback loops.
- **Self-Healing Test Scripts:** Automatically adapting to UI or data structure changes.
- **Cloud-Native Execution:** Enabling on-demand test orchestration across virtualized environments.

These upgrades will enhance agility, optimize resource utilization, and align QA with the firm's digital transformation roadmap.

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